MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 5755

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Twin triode, miniature, medium Mu, dc amplifier

Outline	---	6-2 (EIA)
Base	---	E9-1
Envelope	---	T6-1/2
Cathode	---	Coated unipotential

Base connections:

Pin No.	---	1	2	3	4	5	6	7	8	9
Element	---	2a	2k	2g	h	h	1g	1k	1a	hct

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Eb	Ec	Ehk	Pp/a	TE	Ⓑ Alt
Unit:	V	Vdc	Vdc	v	W	°C	ft
Maximum:	6.9 13.8	250	---	75	1.0	65	Note 1
Minimum:	5.7 11.4	---	---	---	---	---	---
TEST CONDITIONS:	6.3	180	0	---	---	---	---

GENERAL:

Qualification - Required (see note 2)

Ⓑ denotes changes

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1							
1301	Heater current		0.65	II	If	330	390	mA
1256	Electrode current (anode)	See note 3	0.65	II	Ib	1.3	2.9	mAdc
1306	Transconductance (1)	See note 3	0.65	II	Sm	1,100	2,000	µmhos
1306	Transconductance (2)	Ef = 5.7 V (see note 3)	0.65	II	ΔSm Ef	---	15	%
---	Balance	See notes 4 and 5	0.65	II	ΔE1c E2c	---	±0.3	Vdc
1316	Amplification factor	Eb = 110 Vdc; Ec/Ib = 0.150 mAdc (see notes 3 and 4)	0.65	II	Mu	63	77	---
(B) 1266	Total grid current	Eb = 110 Vdc; Ec/Ib = 0.150 mAdc (see notes 3, 4, and 6)	0.65	II	Ic	-1	+0.2	nAdc
1201	Short and discontinuity detection		0.4	II	---	---	---	---
	Quality conformance inspection, part 2							
---	Drift test	See notes 4, 7, and 8	---	---	ΔEc	---	5	mVdc
---	Electrical stability	See notes 4, 7, and 9	---	---	ΔEc	---	2	mVdc
---	Mechanical stability	See notes 4, 7, and 10	---	---	ΔEc	---	25	mVdc
1031	Low-frequency vibration	Rp = 10,000 ohms; Ec = -1.0 Vdc (see note 11)	---	---	Ep	---	20	mVdc
1336	Heater-cathode leakage	See note 11	---	---	Ihk	---	20	µAdc
(B) 1211	Insulation of electrodes		---	---	---	---	---	---
(B) 1121	Base strain		---	---	---	---	---	---
(B) 2126	Glass strain		4.0	I	---	---	---	---
1105	Permanence of marking		---	---	---	---	---	---
	Quality conformance inspection, part 3							
1501	Intermittent life test	Group A; Eb = 120 to 135 Vdc; Ec/Ib = 0.140 to 0.160 mAdc; TA = room	---	---	---	---	---	---
---	Intermittent life-test end point (500 hours)	Transconductance (1) (see note 3)	---	---	Sm	1,000	---	µmhos

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NOTES:

1. See "Reduced pressure (altitude) rating", and altitude, maximum peak voltage in the basic document.
2. Two sets of six tubes shall be submitted for qualification approval tests, one set for the tests not requiring aging, and the second set for the tests requiring aging (see note 4). These samples shall not be submitted to stabilization tests immediately preceding qualification approval tests.
3. Test each unit separately, but with test voltages supplied to both units.
4. The tubes shall be aged for at least 150 hours, in a circuit equivalent to that shown on figure 1, prior to testing. The drift test shall precede the other tests requiring aging.
5. With E1b = E2b = 110 Vdc and I1b = I2b = 0.150 mAdc; $\Delta E1c/E2c = E1c-E2c$.
6. A negative-grid current shall be considered as flowing in such a direction that the insertion of a resistance in series with the grid results in an increase in anode current.
7. At the start of the test an initial adjustment of grid bias shall be made to balance the anode currents in the two units within 0.001 mAdc. ΔEc is defined as the steady-state change in grid bias that is required to return to the initial anode current condition after operation under test conditions for a specified period of time. No adjustments shall be made between observations.
8. This test shall be made on at least a 15 percent sampling basis in a circuit equivalent to that shown on figure 1. The test period shall be a minimum of 16 hours, observations may be made at the start and finish of the final 7-hour period. The limit specified shall apply to the arithmetic average for the sample, disregarding direction of change.
9. Upon completion of the drift test ΔEc shall be observed for an operating period of 5 minutes in a circuit equivalent to that shown on figure 1. The limit specified shall apply to individual tubes. Any failure in the samples shall be cause for rejection of the lot.
10. Upon completion of the electrical stability test, each tube in the sample shall be tapped once in each of six directions with a shock between 400 and 600 G's of approximately 1-millisecond duration. The tapper used shall be equivalent to that shown on figure 2. Ec, when measured in a circuit equivalent to figure 1, shall be interpreted as the maximum variation from the initial setting that takes place following each of the six shocks. The limit specified shall apply to the arithmetic average for the sample, disregarding direction of change.
11. Test with units connected in parallel.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Army - MU, MI
Air Force - 17
DSA - ES

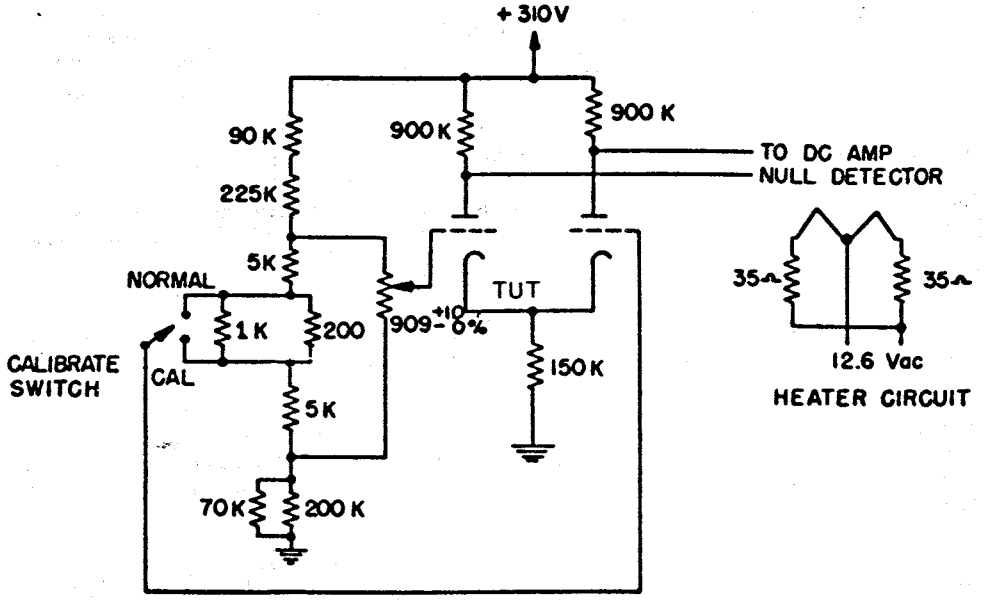
User activities:
Army - WC
Navy - AS, MC, CG, SH
Air Force - 19

Preparing activity: Navy - EC

Agent: DSA - ES

(Project 5960-2480)

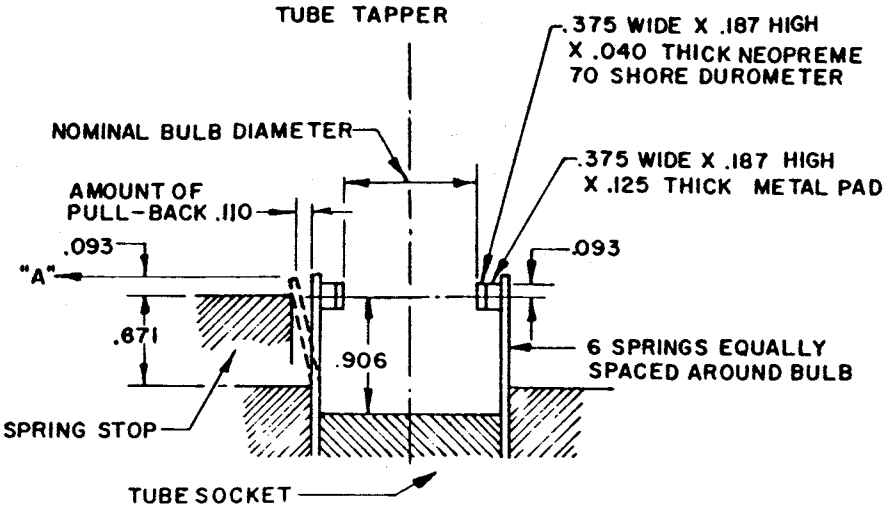
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NOTES:
a. Tolerance on resistors to be 1 percent.
b. Adequate by-pass capacitors should be provided to prevent parasitic oscillations.
c. Approved alternate test: The drop across a 20 megohm resistance in series with the grid shall be between +4.0 and -20.0 mVdc. Ib limits correlated with the grid current limits may be used.

FIGURE 1. Aging and test circuit.

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INCHES	MM
.040	1.03
.093	2.36
.110	2.79
.125	3.18
.187	4.75
.375	9.53
.671	17.04
.906	23.01

NOTES:
a. Approx. width of spring = .375(9.53 mm) x .017(.43 mm) thick phosphor bronze.
b. 'A' - Adjust spring to 7-3/4 ± 1/4 lbs/ in. measured at point indicated.

FIGURE 2. Tube tapper.

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